The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Big Data Analytics), programare as follows

PEO No.	Education Objective
PEO 1	Develop in depth understanding of the key technologies in data engineering, data science and business analytics.
PEO 2	Practice problem analysis and decision-making using machine learning techniques.
PEO 3	Gain practical, hands-on experience with statistics, programming languages and big data tools through coursework and applied research experiences.

Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Big Data Analytics), graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop and implement big data analysis strategies based on theoretical principles, ethical considerations, and detailed knowledge of the underlying data.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analyzing data for decision-making.

1. Course Plan

1.1 Primary Information

Course Name	:	Algorithms and Data Structures for Big Data Lab [BDE 5151]

L-T-P-C	:	0-0-3-1
Contact Hours	:	36 Hours
Pre-requisite	:	Programming with Python or C
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No. of Contact Hours	Program Outcomes (POs)	BL
CO1	Implementation of linked lists, stack and queues	15	PO3	5
CO2	Implementation of binary search tree, sorting and searching, dictionary and Hash Table.	12	PO4	5
CO3	Apply graphs and shortest path techniques for the practical problems.	9	PO4	5

1.3 Assessment Plan

Components	Lab test	Flexible assessments	End semester exam (ESE) / Makeup exam

Duration	90 minutes	Two to three months	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Design	Design	Design
Pattern	Answer all the questions. Maximum marks 30	Assignment: Develop applications using various data structures and different design techniques	Answer all the questions. Maximum marks 50
Schedule	As per academic calendar.	End of Ocotber	As per academic calendar.
Topics covered	Linked List, Stack, Queue, Tree, Searching & Sorting, Hash tables, Graphs	Linked List, Stack, Queue, Tree, Sorting, Hash tables	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.

1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---

L1	Linked List : Implementing Single Linked List	CO1
L2	Linked List : Implementing Double Linked List	CO1
L3	Linked List : Application development using linked lists	CO1
L4	Stack : Implementation and applications of Stack	CO1
L5	Queue: Implementations of Queue	CO1
L6	Tree - Implementation and applications of Tree	CO2
L7	Applications using different search and sorting techniques	CO2
L8	Applications using different search and sorting techniques	CO2
L9	Hash Table - Applications	CO2
LT	Lab Test	CO1, CO2, CO3
L10	Graph representation using list and matrix method	CO3
L11	Graph traversal	CO3
L12	Graph: Shortest path technique	CO3

1.5 References

1. Introduction to Algorithms - Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest. MIT Press.
2. Data Structures and Algorithms - Aho, Hopcroft and Ulmann. Pearson Publishers.
3. Data Structures and Algorithms in Python - Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser. John Wiley & Sons.
4. Data Streams: Algorithms and Applications - S. Muthukrishnan. Foundations and Trends in Theoretical Computer Science archive, Volume 1 Issue 2, August 2005, Pages 117 – 236.

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.

1.7 Course Timetable